

ones created by *mod_env*. Environment variables are provided to CGI scripts as native system environment variables (https://httpd.apache.org/docs/2.4/mod/mod_env.html).

Benchmarking with ApacheBench

We will do a simple Apache benchmarking test for the processing time on *mod_cgi* and *mod_perl*. Values may differ, depending on the CPU/memory configuration. I used 2GB RAM with a single core AMD 2GHz CPU.

ApacheBench (AB) is a load testing utility from Apache that comes along with Apache Dev tools. Shown below is a command that works on any GNU/Linux flavour to check if AB already exists.

```
bejoy@linuxforu:~> rpm -q --whatprovides `which ab`
apache2-utils-2.4.33-lp150.2.6.1.x86_64
```

If the package does not exist, we can install the corresponding development tools.

For Red Hat and Amazon Linux flavours, type:

```
yum -y install httpd-tools
```

For SUSE Linux, the command is:

```
zypper -n install apache2-utils
```

For Ubuntu flavours, give the following command:

```
apt-get -y install apache2-utils
```

For AB testing for *mod_cgi*, use the following code:

```
bejoy@linuxforu:~> ab -n 1000 -c 100 http://IPAddress/cgi-bin/printenv
```

```
This is ApacheBench, Version 2.3 <$Revision: 1826891 $>
Copyright 1996 Adam Twiss, Zeus Technology Ltd, http://www.zeustech.net/
Licensed to The Apache Software Foundation, http://www.apache.org/
```

```
Benchmarking X.X.X.X (be patient)
Completed 100 requests
Completed 200 requests
Completed 300 requests
Completed 400 requests
Completed 500 requests
Completed 600 requests
Completed 700 requests
Completed 800 requests
Completed 900 requests
Completed 1000 requests
Finished 1000 requests
```

```
Server Software:      Apache/2.4.33
Server Hostname:     X.X.X.X
Server Port:         80
```

```
Document Path:       /cgi-bin/printenv
Document Length:     705 bytes
```

```
Concurrency Level:    100
Time taken for tests: 12.016 seconds
Complete requests:    1000
Failed requests:       0
Total transferred:    911021 bytes
HTML transferred:     705000 bytes
Requests per second:  83.22 [#/sec] (mean)
Time per request:     1201.591 [ms] (mean)
Time per request:     12.016 [ms] (mean, across all concurrent requests)
Transfer rate:        74.04 [Kbytes/sec] received
```

Connection Times (ms)					
	min	mean	[+/-sd]	median	max
Connect:	0	171	166.9	121	561
Processing:	139	1004	321.9	1071	1913
Waiting:	8	896	315.8	917	1723
Total:	140	1175	248.7	1196	1930

Percentage of the requests served within a certain time (ms)	
50%	1196
66%	1282
75%	1329
80%	1349
90%	1434
95%	1493
98%	1576
99%	1636
100%	1930 (longest request)

You can use the following code for AB testing for *mod_perl*:

```
bejoy@linuxforu:~> ab -n 1000 -c 100 http://IPAddress/cgi-perl/printenv
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Licensed to The Apache Software Foundation, http://www.apache.org/
```

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Benchmarking X.X.X.X (be patient)
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